磁声波.

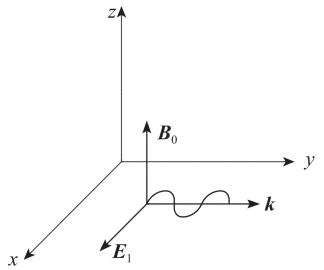


图 4-52　与 $\boldsymbol{B}_0$ 成直角传播的磁声波的几何图

运动方程:

$$Mn_0\frac{\partial u}{\partial t} = en_0(\vec{E} - \vec{v_i}\times\vec{B_0}) - \gamma_i K T_i \nabla n_1$$

$$\begin{cases} -iMn_0\omega v_{ix} = en_0E_x + ev_{iy}B_0 \\ -iMn_0\omega v_{iy} = -ev_{ix}B_0 - ik\gamma_i K T_i n_1 \end{cases}$$

$$\Rightarrow \quad \begin{cases} v_{ix} = \dfrac{ie}{M\omega}(E_x + v_{iy}B_0) \quad ① \text{ 代入 } ② \\ v_{iy} = \dfrac{-ie}{M\omega}(v_{ix}B_0) + \dfrac{k^2}{\omega}\dfrac{\gamma_i K T_i}{M} v_{iy} \end{cases}$$

$$\Rightarrow \quad \begin{cases} v_{ix} = \dfrac{ie}{M\omega}E_x + \dfrac{\Omega_c^2}{\omega^2}\dfrac{1}{(1-A_1)} v_{ix} \\ v_{iy}(1-A_1) = \dfrac{-i\Omega_c}{\omega} v_{ix} . \end{cases}$$

令 $A_1 = \dfrac{k^2\gamma_i K T_i}{\omega^2 M}$　　$\Omega_c = \dfrac{eB_0}{M}$.

$$v_{ix} = \left[1 - \frac{\Omega_c^2}{\omega^2(1-A_1)}\right] = \frac{ie}{M\omega}E_x$$

Maxwell eq:

$$(\omega^2 - c^2k^2)\vec{E}_1 = -4\pi i\omega \vec{j}_1$$

x 方向:

$$(\omega^2 - c^2k^2)E_x = -4\pi i\omega n_0 e(v_{ix} - v_{ex}).$$

同理，对电子，$M \to m$，$e \to -e$，$\Omega_c \to -\omega_c$

$$v_{ex}\left[1 - \frac{\omega_c^2}{\omega^2(1-A_2)}\right] = -\frac{ie}{m\omega}E_x,$$

其中，$A_2 \equiv \frac{k^2\gamma_e KT_e}{\omega^2 m}$

$\frac{\omega_c^2}{\omega^2} \gg A_2 = \frac{k^2\gamma_e KT_e}{\omega^2 m}$.

$$v_{ex} = \frac{-ie}{m\omega}\left(\frac{1-A_2}{1-A_2-\omega_c^2/\omega^2}\right)E_x$$

$\omega_c^2 \gg \frac{k^2\gamma_e KT_e}{m}$

$$= \frac{ie}{m\omega}\frac{\omega^2}{\omega_c^2}(1-A_2)E_x$$

$\frac{\omega_c^2}{k^2} \gg \frac{\gamma_e KT_e}{m}$.

$$= \frac{ie}{m\omega}\frac{\omega^2}{\omega_c^2}\left(1 - \frac{k^2}{\omega^2}\frac{\gamma_e KT_e}{m}\right)E_x.$$

$v_{ep} \gg v_{th}$

近似 $\frac{\omega^2}{k^2} \ll \frac{\gamma_e KT_e}{m} = v_{th}^2$

相速度远小于热运动速度.

$$\simeq \frac{-ie}{m\omega}\frac{\omega^2}{\omega_c^2}\frac{k^2}{\omega^2}\frac{\gamma_e KT_e}{m}E_x.$$

$$= \frac{-ik^2}{\omega B_0^2}\frac{\gamma_e KT_e}{e}E_x$$

$$(\omega^2 - c^2k^2)E_x = -4\pi i\omega n_0 e\left[\frac{-ie\omega^2}{M\omega\omega_c}(1-A)E_x + \frac{ik^2}{\omega^2 B_0^2}\frac{\gamma_e KT_e}{e}E_x\right]$$

Def: $\Omega_p \equiv \frac{4\pi n_0 e^2}{M}$，$v_A^2 \equiv \frac{c^2 B_0^2}{4\pi n_0 M}$，代入:

$$(\omega^2 - c^2k^2) = \frac{-\Omega_p^2}{\Omega_0^2}\omega^2(1-A_1) + \frac{k^2c^2}{v_A^2}\frac{\gamma_e KT_e}{M}$$

$$\omega^2\left[1+\frac{\Omega_p^2}{\Omega_c^2}(1-A_1)\right] = c^2k^2\left(1+\frac{\gamma_e KT_e}{Mv_A^2}\right)$$

$$\therefore\quad \omega^2\left(1+\frac{c^2}{v_A^2}\right) - \omega^2\frac{c^2}{v_A^2}\frac{k^2}{\omega^2}\frac{\gamma_i KT_e}{M} = c^2k^2\left(1+\frac{\gamma_e KT_e}{Mv_A^2}\right)$$

$$\omega^2\left(1+\frac{c^2}{v_A^2}\right) = c^2k^2\left(1+\frac{\gamma_e KT_e + \gamma_i KT_i}{Mv_A^2}\right)$$

$$v_S^2 \equiv \frac{\gamma_e KT_e + \gamma_i KT_i}{M}$$

声速.

$$\frac{\omega^2}{k^2} = c^2\frac{v_S^2+v_A^2}{c^2+v_A^2}$$

- $B_0 \to 0$, $v_A \equiv \frac{c^2B_0^2}{4\pi n_0 M} \to 0$ $\quad \frac{\omega^2}{k^2} = v_S^2$.

  无外磁场 只有声波.

- $kT \to 0$, $v_S \to 0$. $\quad \frac{\omega^2}{k^2} = \frac{c^2v_A^2}{c^2+v_A^2}$